



An Integrated Framework for Service Engineering: A Case Study in the Financial Services Industry

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ABSTRACT

Service engineering, the application of engineering disciplines to develop service-oriented enterprise systems (service systems) with predictable results, has faced a brand new array of challenges in recent years. Existing development approaches that might help addressing these challenges are scattered in separate research fields or in different units of an organization, just like the “blind men and the elephant.” This article presents a 3-year action research case study with a Fortune 50 company in the financial services industry, validating an integrated service-oriented business-IT alignment framework, called the BITAM-SOA Framework, to shed light on complex service engineering issues, including the interplay between business-IT alignment approaches (alignment via architecture, via governance and via communication) and Service Oriented Architecture adoption as well as methodological challenges of service engineering. The importance of the social dimension of service engineering is illuminated. The results yield lessons learned, managerial insights and integrated methods for effective service engineering.

Categories and Subject Descriptors

D.2 [Software Engineering]; K.6 [Management of computing and Information systems]; K.6.1 [Project and People Management] Systems Development

General Terms

Management, Architecture, Design, Human Factors

Keywords

Service Engineering, Service-Oriented Architecture, Business-IT Alignment, Integration, Multi-Disciplinary Research, Framework

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1. INTRODUCTION

Service engineering has recently taken on a new meaning with a new array of challenges as the service concept, business and technological environments have changed dramatically in the past few years. The service concept has shifted in definition, scope and means—from value-added or intangible kinds of goods (Goods-dominant logic) to an emphasis on value co-creation with customers (Service-dominant logic) [24], from primarily human-to-human interaction to IT-enabled interaction [6], from siloed applications to an enterprise focus [4][22]. Service Engineering is now being seen as part of a new discipline called SSME—Service Science, Management, and Engineering—the application of engineering disciplines to developing service systems with predictable results [16]. From an organizational perspective, a service system refers to “dynamic value co-creation and configuration of resources, including people, organizations, shared information (language, laws, measures, methods), and technology, all connected internally and externally to other service systems by value propositions.” [16] Engineering such a complex service system requires a multidisciplinary strategy and faces many new challenges.

First, service engineering faces the chronic system development problem—business-IT alignment—which has been a top concern of CIOs for the past two decades. Business and IT communities traditionally don’t understand each other well. For instance, a service in a business sense is a provider-client interaction that creates and captures value [12]. Within IT, a service is defined as a self-contained, distributed component with a published interface that supports interoperability, is discoverable, and is dynamically bound [21]. Similarly, “performance” in a business differs from “performance” in IT. Challenges remain regarding how to align IT capabilities to achieve business goals in offering value propositions, for instance to “orchestrate” lower level IT infrastructure services to deliver desired business-level customer services or to engineer IT service performance and innovation that increases firm performance in a fast changing regulatory and competitive environment.

Second, as service-orientation takes center-stage in business competitiveness, enterprise SOA (Service Oriented Architecture) has become the number one technology that major corporations are adopting or considering adopting and is seen as key to

realizing effective service engineering to manage rapid changes [13]. (Hereafter, Enterprise SOA will be referred to just as SOA unless specified otherwise.) Rapid changes in the business and technical environment include: changing customer expectations in the “flat” world, lower entry barriers in the service industry, strict legal requirements such as Sarbanes-Oxley (2002), HIPAA (1996) for the healthcare industry, Basel II (2004) for the banking industry, and sourcing changes due to technological advances, including Software as a Service, Service Grid, and so on. Organizations want dynamic system capabilities that can adapt to such changes quickly and can take advantage of contemporary sourcing models.

The technical tenet of SOA lies in: open standards for interoperability, loose coupling, reuse of shared services, and dynamic “orchestration” of component services.. With a SOA in place, organizations hope to leverage existing systems in a dynamic environment, abstracting the essence of applications into services that can be reassembled quickly into new business solutions [3]. The modularity and loose coupling of “services” and their reuse is attractive as it represents potential savings and flexibility for service system development. Furthermore, interoperability via open standards allows for easier integration of both internal and external resources. However, an enterprise SOA catalyzes many changes—organizational, cultural, business processes as well as the service (and system) design and development process. For instance, it requires architects and developers to step out of their silos, to have a deeper understanding of business models, governance and communication, to conform to enterprise architecture technical standards, to know the existing service building blocks and to create new services for future sharing. It would require individual Lines of Business (LOB) to collaborate with each other, which we will illustrate further in the case study. As a result, although we agree with SOA enthusiasts that it is the most promising technology for Business-IT alignment, we argue that SOA alone cannot achieve the desired advantages—its implementation requires careful business-IT alignment throughout the three layers (Business Model, Business Architecture, IT Infrastructure) and 2 dimensions (Technical and Social) of a business system via architecture, via governance and via communication as depicted in the BITAM-SOA Framework [3] [4] shown in Figure 1.

Understanding this interplay between business-IT alignment approaches and SOA adoption is critical in developing service-oriented methods for effective service engineering in today’s business context, which is our ultimate research goal. We are particularly interested in the “how” of alignment and critical success factors for SOA adoption with respect to business-IT alignment for service system development. As our research questions cannot be studied outside real world contexts and we hypothesize that existing alignment methods need to be modified for service orientation, we have chosen a case study methodology [25] which involves action research [15][20][23]. The BITAM-SOA Framework integrates three business-IT alignment approaches with the SOA paradigm, serving as a blueprint for addressing the two afore-mentioned challenges of service engineering. Resulting from the action research, the Framework’s instantiated process model called “Service

Engineering Schematic” [3] outlines an architecture-centric approach for tying together alignment concerns from different disciplines and provides a service-oriented “value-driven” system design process linking different functions/modules in an organization, from architecture alignment, service design, strategic performance management, service orchestration, service level management, IT service management, to eventual service delivery and support that realizes value propositions to customers. An integrated method called the Service Engineering Method (SEM) is formulated based on the Service Engineering Schematic.

The company in our study—hereafter called BFE (Big Financial Enterprise)—is a diversified financial services company providing banking, insurance, investments, mortgage and consumer finance for more than 23 million customers through 6,000 locations, the internet and other distribution channels across North America and the world.

In the next section, the action research case study method will be described. Section 3 presents the research framework and the study itself. Section 4 reports the results and lesson learned, and Section 5 concludes.

2. RESEARCH METHOD

A case study research method [25] is an empirical inquiry aimed at revealing aspects of a contemporary phenomenon inseparable from its real-life context, and thus, difficult to replicate in a laboratory environment. Our motivation for using this method rests on the following: a) exploratory case studies are particularly appropriate for initial evaluation as they allow the course of the study to be adjusted along the way to account for what is learned; b) case studies are also the most satisfactory approach when there are many variables of interest and few data points and where resources do not permit enough replications to isolate individual variables; and c) case studies are well-suited to learn about phenomena in real-life contexts rather than in isolation, and it is the organizational context (i.e., the social dimension) of service engineering, business-IT alignment and SOA adoption that is of great interest. We also use a particular form of action research called *collaborative practice research* [10][17] in this case study. This has involved collaboration between researchers and practitioners from the fall of 2005 to the end of 2007. These activities include alignment via architecture utilizing a modified version of the Architecture Tradeoff Analysis Method (ATAM) [1][7] which was applied through several iterations to generate the service engineering method, SEM. The authors are collectively: creators of the method, involved in the development of the new method, and active SOA adopters.

Our research is based on two questions: 1) How do business-IT approaches interplay with SOA adoption for service engineering? 2) How can existing business-IT alignment approaches be extended and integrated with the SOA paradigm for effective service engineering? By answering these questions we hope to undercover methodological voids and identify practical guidelines.

Based on the research questions, the research site has to be representative of companies facing the two afore-mentioned

service engineering challenges. BFE was chosen as it met the following criteria for study: 1) in a service-intensive and customer-oriented industry; 2) facing intense competition; 3) has both software and enterprise architecture practice; 4) has an established IT governance practice; 5) has observable communication channels; and 6) was adopting enterprise SOA. The research was initiated when BFE embarked on a major service engineering program in 2005. The practitioners and the researchers came together when BFE was applying an architectural analysis and evaluation method—the ATAM [1]—for discovering in business-IT alignment and SOA adoption risks. One researcher is an author of the ATAM method, which was modified through several iterations in this study for the service system environment. One researcher authored the BITAM-SOA framework used in this study. One researcher led the ATAM workshops and participated in the SOA adoption. The research team has first-hand knowledge of the initiation, planning and execution of the methods and worked with practitioners to improve the methods. The team has access to documentation of meetings, workshops and program files associated with various methods performed for service engineering. Data were collected for analysis from documentary sources, workshops, meeting notes, email communications and dialogues with practitioners.

3. ACTION RESEARCH PROCESS AND RESULTS

3.1 Case Study Setting

BFE, a Fortune 50 company with 150,000 employees, provides financial services and products in an extremely competitive environment and is strategically focused on retaining existing customer relationships and expanding market share by both increasing its customer base and increasing the number and type of financial products/services to each of their 23 million customers. The enterprise is supported by an in-house IT and systems development organization (with 7,000 employees) which has applied aspects of SOA (among other strategies) to increase flexibility in delivery for business needs, improve time to market, and enable more seamless operation among LOBs. The case study focuses on a joint business and technology strategic initiative across three lines of business (involving over 50,000 employees).

3.2 The Framework

The framework for this study is the BITAM-SOA framework as shown in Figure 1. Detailed descriptions of the framework can be found in [4], and will not be repeated here due to space limitations. The study will followed two threads. First we will examine the three alignment approaches focusing on different aspects of an organization: Alignment via Communication, Alignment via Architecture (including software architecture and enterprise architecture), and Alignment via Governance, including business performance management (BPM) and IT service management (ITSM). Two alignment operations include vertical alignment (between layers) and horizontal integration (at

each layer). The second study thread is on SOA adoption including service-oriented methods. The analysis follows the seven principles of the Framework: 1) customer-centric value proposition, 2) layered abstraction and encapsulation, 3) enterprise cross-functional integration (strategy, process, resources), 4) Enterprise SOA paradigm, 5) integrated metadata management, 6) integration of business performance and IT governance measures, 7) explicit attention to the social dimension. The subsequent analysis will focus on the modules (e.g., functions) and the linking of these modules as depicted in Service Engineering Schematic in Figure 2. The modules were designed based the integration of business-IT alignment approaches and the SOA paradigm.

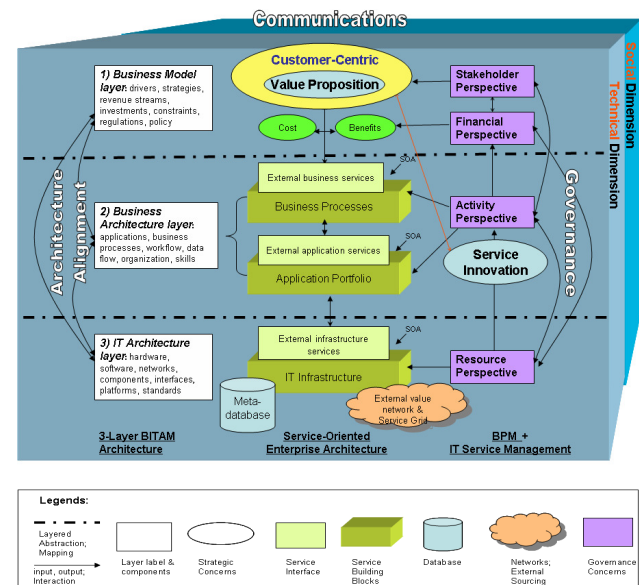


Figure 1. BITAM-SOA Framework

3.3 BFE's Strategic Initiative

BFE embarked on a major initiative to improve customer centricity, time to market, and efficiency. To achieve these goals, BFE decided to focus on creating a standard suite of products/services across the business built from a common architecture.

This initiative was in response to the realization that the state of business and technology implementation had produced a “vicious cycle” of complexity. This complexity was recognized as existing in each major element of the enterprise architecture (products/services, documents, process, and technology) and in their interactions. Complexity in each element drove second order complexity in the others, producing a multiplier effect. Past attempts to simplify any single element independent of the others yielded only marginal benefits and were difficult to sustain and the unresolved complexity would, over time, diminish the effects of the isolated improvement. BFE reached the conclusion that *all elements* would have to be addressed and aligned to realize and sustain the full benefits of simplification.

3.4 Business-IT Alignment in BFE

BFE's existing efforts in alignment addressed all three aspects shown in the Framework in Figure 1:

Alignment via Architecture: the IT organization maintained a “strategic roadmap” to help prepare the LOBs for the “next wave” of technology within the industry, recognizing that innovation and improved time to market were critical competitive success factors. The roadmap and its associated Living Target Architecture (LTA) detailed the key architectural and technical principles guiding the development of new systems and services, and evolution or replacement of existing systems and services. The technical dimension of alignment was guided by the IT architecture vision which detailed key approaches and technologies to enable alignment across functions and systems. These approaches and technologies included pattern-oriented design, component-based development, message-oriented middleware, web-enabled applications, and SOA. The vision of the IT architecture organization explicitly recognized SOA as a key element with the intent that the organization would acquire or build services to perform specific business functions.

Alignment via Governance: BFE used a scorecard approach [14] to measuring business satisfaction with IT services and performance. One of the scorecard measures was “LTA's are relevant to the business and clearly state so via their roadmaps, and technology projects are aligned to the roadmaps”.

Alignment via Communication: A previous version of the strategic roadmap (encompassing both business and IT) identified six major strategic business-IT programs which were being managed under an umbrella program office. In this way the programs were managed separately but with senior leadership oversight to maintain alignment across the initiatives. While overall strategic alignment was maintained, the management of each program as a separate and distinct effort presented ongoing challenges in synchronizing schedules, designs, and implementations to ensure that the end state would provide business users with a well integrated set of systems in time to meet changing market needs.

3.5 SOA adoption in BFE

Enterprise SOA adoption at BFE initially followed a bottom-up evolutionary process. Earlier SOA implementation efforts had proved successful in meeting the needs of individual LOBs. In the previous 3-4 years there had been several successful initiatives which developed service frameworks, supporting processes, and specific services. For SOA adoption at the enterprise scope, the architectural maturity of BFE was assessed. A third party architecture review had validated that the services approach was maturing and included solutions to the most common pitfalls observed in SOA implementations. The review concluded that BFE had a strong approach for enabling the next generation of business services within the enterprise. The individual SOA implementations had matured to the point where they were providing reliable, managed delivery to a growing user base at high volumes of demand. Each LOB had also matured their customer co-creation model and a pipeline of new services definition and development was growing in each area.

3.6 Conducting the Action Research

The desire to integrate these three approaches and the SOA paradigm resulted in a proposed program structure change at BFE, which was designed to improve the alignment of the many stakeholders both within and between the business and IT organizations while continuing the overall course set by the previously developed strategic roadmap. As part of the effort to fulfill the scorecard goal that “LTA's are relevant to the business and clearly state so via their roadmaps, and technology projects are aligned to the roadmaps,” IT leadership committed to complete design assessments for each release to ensure services were in alignment with the documented risks and trade-offs. Previously, a number of reviews, evaluations, and assessments had been performed on service development efforts and on components of the program under development. These ranged from brief, informal peer reviews by LOBs and enterprise architects to formal ATAM reviews (on two of the component systems) performed by third party consultants. While existing review methods were focused on evaluating the architecture and functional design of component systems, the senior technical leadership determined that the goal of design assessments was to conduct an “*end to end*” evaluation of the integrated service systems architecture. This was the initial goal of our action research.

The more formal approaches investigated for the project goals included: ATAM, ARID [8], ADR [18], and DOD PDR and CDR [11]. While each approach had its advantages, each also had its own obstacles, which made it unsuitable to adopt wholesale in BFE environment. Therefore it was initially decided to employ a scenario-based approach modified from the ATAM, called the BITAM [5]. The research team followed the action research cycle [10][17] in developing the eventual method for BFE, the Service Engineering Method (SEM), which will be introduced in Section 4 and detailed in another paper.

First Iteration- ATAM Modified: In the initial round of review planning and execution the ATAM was modified to extend the pre-work, leveraging existing program architects and a select group of technical leaders to seed the scenarios and the utility tree. In the formal review session, conducted in October 2006, the highest priority scenarios were presented and architects of component systems walked through logical and physical views and described availability and performance approaches and tactics. Risks and action items were documented for follow-on. While the ATAM was a useful exercise, particularly in encouraging the architects to walk through their documentation and plans, there were challenges associated with the lack of familiarity with the method. Also, organizational separations between business and technology made it difficult to create a collaborative, participatory environment for the reviews.

Second Iteration- More Training on ATAM: Given the experience with the first review, the team made further refinements to the approach in preparation for a second review session held in December. Between the first and second reviews the team sought training in applying the ATAM methods and the lead practitioner obtained ATAM evaluator certification. With the stronger facilitation tactics learned in the ATAM training the review team was able to present several more scenarios and capture more detailed and specific architectural drivers, tactics,

tradeoffs, and risks. These findings were distilled into six major risk themes which were presented to senior program leadership, along with more detailed documentation of individual findings.

Third Iteration- BITAM was Employed: Both reviews highlighted risks around business-IT alignment. Although, much effort had gone into aligning during the business process modeling (Layer 1 and Layer 2) activities, the architecture and implementation had diverged later in the process. In part this was due to ambiguity and high levels of uncertainty in the quality attribute requirements (primarily with performance and availability) and their interpretation. IT had accepted non-specific requirements and interpreted them as a shared business-IT understanding of high levels of uncertainty. Subsequently the architects viewed their designs as expressions of what was *technically feasible* in each attribute area. The business had a different view, expecting high levels of performance and availability that were seen as *contracts* for IT to deliver upon. For example, the business defined availability targets at 99.9% for the entire system. The architects identified a constraint that the network infrastructure was only available at 99.7% and knew that availability would be no better than the “weakest” component involved, therefore the 99.9% target could not be met. Rather than proactively communicating this with the business, IT assumed this was a valid and accepted constraint, and proceeded with implementation. The review sessions were the first time this disconnect was clearly surfaced. IT recognized the need to implement a collaborative process with the business which would yield more concrete and discreet quality attribute requirements. The QAW (Quality Attribute Workshop) method [2], a complimentary method to the ATAM, was leveraged in 2007 and a process was defined to incorporate quality attribute requirements into Requirements Management. A Quality Attribute Requirements Questionnaire was developed to capture information and work sessions were defined to be conducted during detailed requirements and high level design phases.

Fourth Iteration- Service Engineering Schematic Created: In the 4th iteration, additional modifications to the BITAM were made to improve ability of future reviews to target areas most critical to the business. Reviewers recognized that for a large system with diverse stakeholder groups, it may not be easy to capture a view of business drivers that is concise yet comprehensive enough for the evaluation. Although the BITAM was useful, specific linkages between different enterprise elements and service orchestration needed to be explicitly established. The Service Engineering Schematic (Figure 3) was developed as a process model for the extended service-oriented method.

Subsequently, quality attribute scenarios were driven from a prioritized list of 27 high value business functions identified by the business as part of the L2 SLM (Layer 2 Service Level management) module shown in the Schematic. The modified BITAM then integrated with the SOA modeling efforts (detailed in the next section). The ATAM utility tree structure was refined by adding thresholds and trigger events to help guide scenario generation and evaluation. The modified utility tree documented

vital quality attribute requirements and their associated thresholds and constraints within the context of the high value business functions and services.

Finally, a change was made to the BITAM to drive follow-on actions by tying review findings to the program risk and issues management system, which included regular monitoring and governance (to be specified in CAAR module in the Service Engineering Schematic). This was important as previous reviews sometimes yielded important risk findings which were not made sufficiently visible to business and IT leadership.

Closing: As the initial individual SOA implementations matured and were recognized as contributing value to the overall business, BFE adopted a common services framework and governance model across the LOBs.

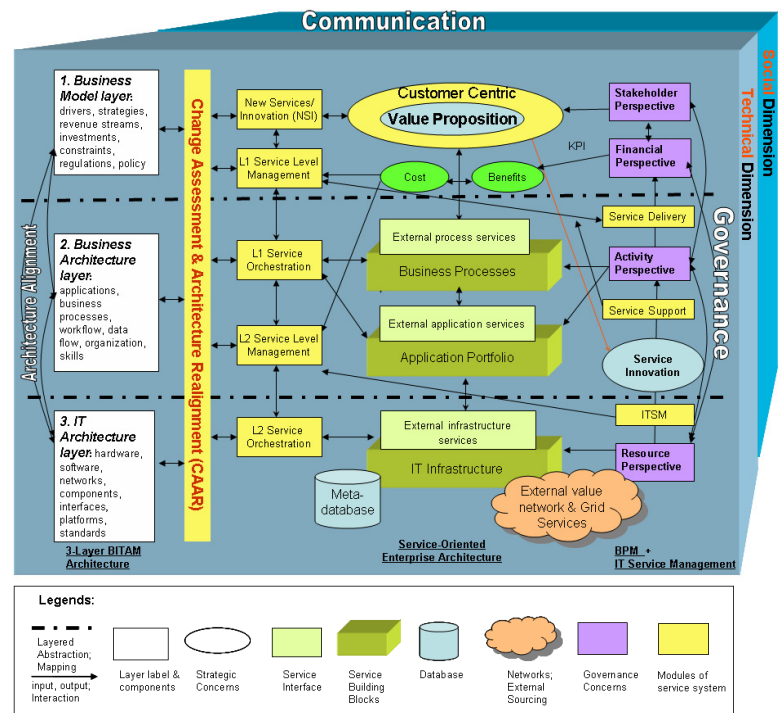


Figure 2. Service Engineering Schematic

The technical architects supporting each LOB collaborated on evaluation of existing implementations and developed a go-forward model to support service development, delivery, and support for BFE as a whole. Existing services were converted to the new framework and new services were developed and delivered in an iterative fashion. Decision criteria for whether existing services should be ported or replaced entirely by different technologies (including new requirements gathering) were developed and each legacy service was dispositioned according to these criteria. Legacy services were retired as the new services were either ported or created anew in response to changing business needs and technologies. Service design and development continued to address services to meet the unique needs of specific LOBs and there was also a growing focus on defining common services (for cross-LOB use) and identifying

services originally developed for one LOB which could be reused, or composed with other services, to meet the need of other LOBs. This was aided by an informative, accessible business services information repository (the meta-database component in the Schematic) which helped identify existing services and their attributes and interfaces. For example, an existing enterprise service to retrieve a customer profile was leveraged in lieu of developing this functionality as part of a new program. This maturing services effort then followed the process for service deployment in the Service Engineering Schematic.

3.7 Benefits Resulted for BFE

BFE has had a double digit profit and annual growth rate during the three-year period of the study and expects only a slightly reduced rate in 2008 despite turbulent market conditions in the financial industry. Three major business objectives of the service engineering program initiative have been achieved: 1) New services for LOBs that had great *time to market* pressure were deployed quickly (e.g., to over 2000 users in 3 months), greatly improving the end user and customer experience which would not have been achievable on this scale and timeframe in a more tightly coupled non-SOA environment. 2) The new SOA based system greatly improved *efficiency*, for example, by managing end-to-end workflows so that participants in the overall process could access their data from the same source and manage their workload queues more efficiently. 3) BFE LOBs developed a “customer centric” architecture approach to the SOA development. One key example is a set of services that were developed to compare two different input sources to closing costs which are driven by regulatory and market factors and often vary slightly. These are an unavoidable part of the financial service but customers were dissatisfied when the presence and amount of the variation was not calculated until late in the process. The new service performed a variation calculation, made the information accessible to the sales representative and incorporated communication into the standard workflow at an earlier stage of the process. These improvements have led to increased customer satisfaction which in turn correlates to increased cross-sell of financial products /services to existing customers.

4. INSIGHTS AND LESSONS LEARNED

In addition to the methodological modification of ATAM to a service engineering method as discussed in Section 3, the following observations were obtained.

4.1 Critical Interplay Between Business-IT Alignment and SOA Adoption

1) *Business-IT alignment and SOA adoption enable each other:* The evolution of both business-IT alignment and SOA within BFE can be seen as two interdependent threads, each necessary for, and driving the growth of the other. Without initial business-IT alignment at the individual LOB level there would not have been support for the initial SOA efforts; without the value delivered by successful LOB SOA implementations, there may

not have been the same level of recognition of the potential for more widespread (cross-LOB) business-IT alignment, which in turn provided support for the unified enterprise SOA framework and implementation. In turn, the focus on service level management (SLM) and service orchestration supported better stakeholder communication and requirements gathering, as well as better project management, scheduling, and execution.

2) *Business-IT alignment in the social dimension is critical for technical dimensions of SOA deployment:* A joint business and technology program structure was defined with representation across all key areas. This group was chartered to establish phase boundaries and define scope. In identifying the senior leadership cadre for the program the sponsors considered not only key functional and leadership characteristics but also the business background and relationships. The program was chartered to support three related LOBs and the program leadership council included a representative assigned from each of the three LOBs as well as a fourth member who was a senior technology executive. This program structure reorganization (driven by the Change Assessment and Architecture Alignment, or CAAR, module) was designed to improve the alignment of the many stakeholders and teams in the business and IT organizations while continuing the course set by the previously developed strategic roadmap. Key to their efforts was the definition of a simple set of business products and services to be implemented as part of the new program while complex products would (for the time being) continue to be delivered via legacy systems and business processes already in place. The approach placed a heavy emphasis on collaboration in the definition of scope, business processes, and high level service level specifications.

4.2 Service Engineering Schematic for Architecture-Centric Methods

The Service Engineering Schematic represents an important output of the action research: a multi-disciplinary approach to service-oriented enterprise system development. Based on the needs of BFE uncovered in the third iteration, the goals of the Schematic were threefold: 1) to outline a set of modules (a module denotes a set of processes, models, methods and techniques for a defined task that may span departmental or organizational boundaries) and the ways in which these modules interact for service design, development and management. 2) to offer a methodological scaffold for integrating existing methods in distinct fields for service engineering. 3) to serve as a blueprint for developing a service-oriented enterprise system as well as a process model for new service development or service modification while aligning Business and IT.

In BFE’s case, the three approaches for alignment were integrated into the CAAR module across three layers in the Schematic, which is used to assess the scope of impact (cost-benefit, risks, etc.) by change event, primarily by means of architecture analysis. In response to this analysis, related modules are activated to respond to changes. Change events may alter the state of system deployment and alignment among the layers and thus catalyze a realignment. Architecture tradeoff analysis is conducted to ensure that levels of service quality are

maintained according to governance standards, and performance metrics. Architecture strategies are evaluated based on cost-benefit analyses of business drivers [7] which ties with BPM measures that include financial perspectives. As such, this module interacts with Service Level Management and Service Orchestration modules (related to ITSM) to determine what components and services need to be bought, built, or reused, focusing on their external properties. The subsequent impact on Service Delivery and Support modules on both social and technical dimensions then needs to be assessed.

The development of the CAAR module in BFE allows the opportunity for our action research as existing architecture analysis methods do not readily apply to the enterprise scope and do not explicitly consider the service-orientation adopted by BFE. Our research tailored a method to fit BFE's culture and project goals. The change is top-down: with the recognition of the multiplier effect of complexities in both the business and IT architectures, the decision was made to more closely align the program efforts by integrating a key set of deliverables into one unified program with an initial phase to implement a set of lower complexity business processes. This initial phase incorporated processes, products, and services from multiple LOBs within the organization, centered the new thinking around a common architecture, and tested the scalability of the new environments.

4.3 Service Engineering: How?

The “how” of service engineering (e.g., design, orchestration, SLM, service support and delivery) in BFE followed a “top-down” approach premised on the alignment of both technical and social dimensions—a point repeated found in our study.

As shown in Figure 2, services, from larger granularity to smaller, are in hierarchical order in the three layers. The L1 Service Orchestration (L1 SO) module will decompose and map required Layer 1 business services onto course-grained services of workflows, processes and applications (elements of the Business Architecture layer) and compose existing coarse-grained services with business logic for implementing new or modified business services. A heavy early emphasis of BFE was placed on detailed modeling and decomposition of the business processes within the scope of the first phase of the initiative (for 3 LOBs). Layer 1 business services were defined at a high level and described the distinct phases of the consumer lending process. Layer 1 included activities such as “Identify Customer Opportunities” and “Initiate Customer Application” which represented the value chain of the customer interaction. These L1 SO module processes were decomposed into a second level of coarse grained services, workflows, and processes. This effort was led by the business architects with widespread participation from key business stakeholders and review by both business and technology program representatives.

The *top-down decomposition strategy* described above served a key role in effective service engineering in a number of ways. First, it helped all stakeholders better understand the nuances and complexity of the business process. A Layer 1 description in and of itself may seem to describe a simple process, and it often is conceptually simple, however when decomposed to the next level there are numerous sub-processes and workflows that must

be defined and addressed to fulfill the highest level process. For example the first layer description of “Initiate Customer Application” was decomposed into 15 second layer processes for further analysis and decomposition.

Secondly the Layer 2 process descriptions served as the basic unit of work description which could then be assigned to a responsible sub-team. This was key to managing a large program with many business and technical staff collaborating on alignment, design, and implementation. The program was organized into four central delivery teams, each with responsibility for implementing a layer of the technical architecture. Each team was staffed with both business and technical management and subject matter experts who were tasked with working together to design, build, test, and implement the component services. Senior leadership in each team had shared goals and both parties were jointly accountable for meeting the program milestones and objectives. The delivery teams were supported by a single program management office and a centralized test organization responsible for verification and validation. Regular interchange between the teams was facilitated by a senior executive leadership team with both business and IT representation so business and IT functions and staff were aligned in both daily working tasks and ongoing strategic planning and management (see Figure 3). When Layer 2 definition was complete each coarse grained service was assigned to one of the delivery teams. Where this course grained service could not be assigned clearly to one team, the interdependencies were mapped and each affected delivery team was represented in the next phase of requirements activities.

Third, it was the business process modeling (in the L1 SO module) that served as the basis for *meaningful* service level agreement (SLA) creation and service level management (SLM) and reporting between the business and IT organizations. Previous service delivery and support in the organization had focused on SLA definition and management at a more granular level and this often failed to promote business and IT alignment in ongoing SLM activities. While it is simpler to define SLAs at a more granular level and less complex to monitor and report on SLA achievement at the granular level, these reports can fail to provide true indication of the systems' ability to provide value to the business. For example, SLA reports generated on a component by component basis might show superb performance against SLAs for 19 out of 20 components and poor performance for only a single component. At face value, this would indicate good performance. In actuality, if the one component which performed poorly was critical to 95% of business functions the business perception will be that SLA performance was very poor, in spite of the high performance for other components. In recognizing that SLA reports needed to reflect *business* value, a better method was created. For example, the practice now reports *not* on what percent of components were available but on what percent of key business functions were available. The new method was able to reframe SLA creation, monitoring, and reporting following this top-down decomposition strategy and thus improve overall business-IT alignment and service effectiveness. For instance, weekly SLA reports were developed which reflected availability of the high value business functions at the Layer 1 process level. On an ongoing basis, specific scorecard measures were used to reflect the degree to which the

business units felt the SLM reporting and performance supported business functions.

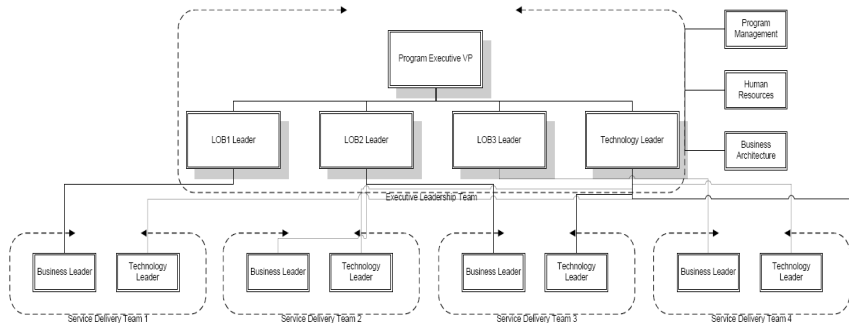


Figure 3: BFE Business-IT Alignment for Service Delivery (Social Dimension)

Additionally, the new method included a *prioritization scheme*. The second layer of the business process model created in this step served as a starting point for further business process decomposition, for modeling, and for a more detailed understanding of the business's requirements. While the second layer business process served as the basis of SLA creation and management, the team recognized that not all second layer processes had the same value and criticality to the business. To make SLM and reporting more manageable and actionable, the research team undertook an effort to prioritize Layer 2 processes and select a subset for monitoring and reporting. End users, business, and IT stakeholder representatives were brought together to review and prioritize the most critical. Of the 82 sub-Layer 2 processes defined, 34 were selected as most critical to customer satisfaction and business success.

These 34 processes were then decomposed into their component technical services and technical details were overlaid to help the service support teams configuring instrumentation and tooling to monitor and report on actual service level performance.

Finally, the Service Support module in Layer 2 ensures that a delivered business service meets customers' expectations and requirements and maintains the Layer 1 SLA. However, selecting, installing, and configuring the appropriate instrumentation and tooling in IT infrastructure level to report on service level performance for the critical business processes was a non-trivial activity. The most granular level of services which were composed to provide an coarse grained service were often deployed on disparate pieces of hardware in data center infrastructure. *Mapping methods* were developed for monitoring the parts and then reporting on service measures for the coarse grained service as a whole.

4.4 Consolidated Lessons Learned

First, *performance measurement alignment*: the focus was initially on measuring the performance of the implemented services and processes as defined in the requirements. The business interest was in measuring the end user experience, which would be greatly impacted by network latency issues. Once again business and IT had to work together to maintain alignment in reporting and understanding the meaning of data points to agree on how best to address problem areas negatively impacting the business experience/business value.

Second, *long-term view on payoff*: for example, developing the capability to report service performance against Layer 2 processes introduced additional effort and cost. However, it was well worth it for truly business critical services with a lengthy (years) lifetime in service.

Third, *caution about decomposition*: while our new method improved upon previous results at business-IT alignment in SLM and Service Support, there were still numerous coordination and social factors which presented barriers to business-IT alignment. Inherent in managing a large team and systems integration effort is the need to decompose functions for implementation and organize work into units that can be accomplished within a manageable team size. This is necessary yet presents its own set of pitfalls. In the decomposition of business processes to more granular services, the business orientation and context which comes with more coarse definitions can erode and the small teams implementing individual services may not understand the business context. This can lead to implementations losing some of the alignment benefits originally designed.

Fourth, *undertake L2 SLM early*: some services are coarse-grained enough at the lowest level to be measured but others must be composed to achieve and understand value. While the L2 SLM and Service Support activities tied this value to individual services, this effort was not complete at the time of individual service design and implementation, and therefore the small implementation teams and the next level integration teams sometimes produced designs and implemented services which could not meet the SLAs for the critical processes. This resulted in later rework and refactoring once the gaps were identified in test and implementation. Future recommendations would be to undertake the L2 SLM activities earlier so that their outputs could be better leveraged in service design, implementation, and delivery.

Fifth, *business inexperience and misunderstandings about SOA*: a lot of marketing hype surrounds SOA adoption efforts in large enterprises like BFE. While the benefits of SOA are well publicized, the tradeoffs inherent in such implementations are not often understood by senior business leaders. At times, there may be unrealistic expectations that SOA will solve all technology problems. For example a by-product of SOA loose-coupling is the overhead associated with messaging; this can have a negative affect on transaction performance. In the quality attribute requirements gathering activities and architecture reviews we often found that business stakeholders expected performance *improvements* over legacy systems. In reality, in

some cases SOA system performance might decrease slightly; a reasonable trade-off for the other benefits (ease of implementation, flexible composition, customer co-creation, etc.) which were primary to the selection of SOA in the first place. Communication and education efforts to help business and IT understand both the benefits and costs of a particular SOA implementation could help stem some of these issues. Additionally, early dialogue and feedback between business and IT during quality attribute requirements activities could help set realistic expectations for system performance.

Finally, *continuous alignment/realignment is required*: while there is an awareness of the criticality of business-IT alignment and while lessons learned from past SOA implementation have helped business and IT leaders focus on early lifecycle activities to promote alignment, our experience has highlighted that it is equally important to reinforce alignment *throughout the lifecycle*, not just in process modeling and requirements gathering but throughout design, implementation, validation, and SLM. Without follow through and ongoing reinforcement, early alignment can dissolve as individual teams work independently to deliver their assigned components and processes.

5. CONCLUSIONS

The action research case study method, which strikes a balance between relevance and rigor [17], is most appropriate to answer our research questions in service engineering even though the generalization of the results beyond the case of BFE may be limited and researchers' biases are embedded in the analyses. The research method we employed allowed us to address the practical problems of BFE while developing new methods and gathering insights for improving service engineering knowledge and practice. The BITAM-SOA framework and its instantiated process model, the Service Engineering Schematic, represent an advancement on the state-of-the-art in service engineering. The present study validates the framework, illustrates the interplay of Business-IT alignment and SOA adoption and provides better understanding of the "how" of service definition, modeling, orchestration, management, and support. The study contributes to the understanding that SOA adoption and Business-IT alignment enable and facilitate each other. Effective service engineering methods must attend to both, and integrated methods that combine alignment approaches and SOA paradigm are needed.

Our action research through 4 iterations over a period of 3 years resulted in a new improved method (Service Engineering Method) which extends the ATAM and BITAM—architectural analysis methods for alignment in the purely technical domain—to the service engineering domain. The scenarios used in the ATAM have been informed by critical business processes which can serve as a binding concept between IT services and business services. Service orchestration and management based on this new alignment method has become easier. The method includes a top-down decomposition strategy, prioritization scheme, and mapping methods among different modules in different layers for service modeling that results in a positive feedback loop for business-IT alignment.

Most importantly, the present study illuminates the importance of the social dimension in every aspect of service engineering. SOA adoption is not just a technical pursuit, it entails major cultural, organizational and process changes and requires additional management capabilities. The results and lessons learned provided by this study are intended to help other organizations improve or better prepare for their service engineering endeavors.

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